

RESEARCH ARTICLE



Intranasal nanoemulsion-based *in-situ* gel containing dexlansoprazole: formulation development, *in-vitro* and *ex-vivo* evaluation

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ABSTRACT

Aim: Dexlansoprazole (DXL) is clinically effective in gastroesophageal reflux and ulcer healing, yet its oral bioavailability is compromised by acid instability, poor aqueous solubility, and extensive first-pass metabolism. To address these limitations, an intranasal nanoemulsion-based *in situ* gel was developed to bypass gastrointestinal and hepatic metabolism and enable rapid systemic absorption *via* the highly vascularized nasal mucosa. Owing to the BCS Class II nature of DXL, this approach was designed to enhance solubility, prolong nasal residence time, and provide sustained drug release.

Methods: DXL-loaded nanoemulsions (NEs) were prepared using Capmul MCM EP as the oil phase, Cremophor EL as the surfactant, and PEG 300 as the co-surfactant by sonication. The NEs were evaluated for various *in vitro* parameters, *in vitro* drug release, and *ex vivo* permeation.

Results: Among all NEs, F3 exhibited optimal characteristics, with a mean droplet size of 40.55 nm, a PDI of 0.374, a zeta potential of -17.70 mV, spherical globules under TEM and the highest cumulative release of 90.02% at 24 h. All NEs displayed significantly enhanced drug release as compared to the drug suspension. Cumulative drug permeation of $90.02 \pm 0.565\%$ for F3 and $80.69 \pm 0.057\%$ for its *in situ* gel (F3NEG4) was recorded, with 5.54- and 5.18-fold higher J_{ss} than the drug suspension, respectively. Additionally, F3 remained physically stable for 3 months at $2-8^{\circ}\text{C}$ and $25 \pm 5^{\circ}\text{C}$.

Conclusion: The developed DXL-loaded NE improved *in vitro* dissolution, indicating potential enhancement of bioavailability, which requires confirmation through comprehensive *in vivo* studies.

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Introduction

Gastroesophageal reflux disease (GERD) is a prevalent chronic gastrointestinal disorder affecting approximately 8–33% of the global population and is characterized by the reflux of gastric contents into the esophagus, leading to both esophageal and extraesophageal symptoms [1]. Proton pump inhibitors (PPIs) remain the mainstay of GERD therapy due to their potent acid-suppressive action.

Dexlansoprazole, a PPI, exerts its therapeutic effect by irreversibly inhibiting the H^+/K^+ -ATPase enzyme responsible for gastric acid secretion. Despite its clinical efficacy, dexlansoprazole suffers from significant formulation and pharmacokinetic limitations. It is a BCS Class II drug with low aqueous solubility, is acid labile, and undergoes extensive first-pass hepatic metabolism following oral administration, resulting in poor and variable bioavailability. Additionally, its short

plasma half-life (1–2 h) necessitates modified-release formulations, which further complicate oral delivery [2].

Intranasal drug delivery presents a promising alternative for drugs such as dexlansoprazole, as it bypasses gastrointestinal degradation and first-pass metabolism while enabling rapid systemic absorption through the highly vascularized nasal mucosa. This route offers the potential for faster onset of action, improved bioavailability, and enhanced patient compliance compared to conventional oral administration [3,4]. However, effective intranasal delivery requires formulation strategies that enhance solubility, permeation, and nasal residence time.

Nanoemulsion-based delivery systems are particularly suitable for intranasal administration of poorly water-soluble drugs due to their nanoscale (20–200 nm) droplet size, high surface area, and ability to improve drug solubilization and mucosal permeation. These properties facilitate rapid and efficient drug absorption

across the nasal epithelium while maintaining physical stability [5,6].

To further overcome rapid mucociliary clearance, the incorporation of *in situ* gelling polymers offers a viable strategy to prolong nasal residence time. Ion-activated polymers such as gellan gum undergo sol-gel transition upon contact with nasal ions, forming a mucoadhesive gel matrix that sustains drug release and enhances mucosal contact. Their major advantage lies in ease of administration in liquid form, improved patient comfort, precise dosing, and sustained residence time at the administration site [7].

Although several formulation strategies for dexlansoprazole—such as floating tablets, micro balloons, gastro-retentive gels, and extended-release oral systems—have been previously reported, these approaches remain confined to the gastrointestinal route and therefore do not address the drug's inherent limitations of acid lability and extensive first-pass hepatic metabolism [8–10]. Consequently, improvement in systemic bioavailability using these oral platforms remains limited.

In contrast, the present work introduces, for the first time, an intranasal nanoemulsion-based *in situ* gelling system for dexlansoprazole, representing a clear departure from previously reported oral and gastro-retentive formulations. The novelty of this approach lies not only in the change of administration route but also in the integration of two complementary delivery strategies: (i) a nanoemulsion system to enhance solubility and transnasal permeation of this BCS Class II, acid-labile drug, and (ii) a gellan gum-based ion-activated *in situ* gel to prolong nasal residence time and regulate drug release.

This dual-platform system enables bypass of first-pass metabolism, protects dexlansoprazole from acidic degradation, and facilitates rapid and sustained systemic absorption via the nasal mucosa—advantages that are not achievable with existing oral formulations. To the best of our knowledge, such a nanoemulsion-*in situ* gel-based intranasal delivery strategy for dexlansoprazole has not been previously reported. Therefore, the proposed formulation constitutes a novel and innovative approach with distinct mechanistic and therapeutic advantages over conventional oral and gastro-retentive systems.

Materials

Dexlansoprazole was procured from MSN Laboratories (Hyderabad, India). Solvent (methanol), surfactants (Tween 20, Tween 80, Span 20, Brij 35), co-surfactants (PEG-300), and oils (oleic acid, ethyl oleate, castor oil) were obtained as excipient samples from Alpha Chem

Pvt. Ltd., Mumbai. Potassium dihydrogen orthophosphate, sodium dihydrogen orthophosphate, sodium chloride, and co-surfactants (propylene glycol, PEG-400) were obtained as samples from Hi-Media Laboratories Pvt. Ltd., Mumbai. Cremophor EL and Cremophor RH-40 were received as gift samples from Trans-Chem Corporation, Mumbai. Capmul MCM was obtained as a gift sample from IMCD India Pvt. Ltd., Mumbai.

Methods

Screening of oil, surfactants, and cosurfactants

An excess amount of DEX (300 mg) was added to glass vials containing 1 ml of various oils (oleic acid, peanut oil, Capmul MCM EP, castor oil, ethyl oleate, and olive oil), co-surfactants (PEG-300, PEG-400), and surfactants (Tween 20, Cremophor EL, Cremophor RH-40, Span 80, Tween 80, and Brij 35). The vials were placed in an orbital shaker (Remi CM-101, India) at 37°C for 48 h under constant agitation at 3500 rpm. After equilibration, the samples were centrifuged, and the concentration of DEX in the supernatant was measured using a UV-visible spectrophotometer at 282 nm. The oil, surfactant, and co-surfactant exhibiting the highest solubility of DEX were selected to construct pseudo-ternary phase diagrams for finalizing the formulation components based on the higher nanoemulsion region [11].

Construction of the pseudo-ternary phase diagram (PTD)

Pseudo-ternary phase diagrams were constructed using Capmul MCM EP as the oil phase, Cremophor EL as the surfactant, and PEG-300 as the co-surfactant using the aqueous phase titration method. An optimal surfactant-to-co-surfactant ratio providing the largest nanoemulsion region was determined from the pseudo-ternary phase diagrams.

Briefly, for nanoemulsion formation, the S_{mix} (surfactant and co-surfactant) was mixed with the oil phase, and distilled water was gradually titrated into the mixture. The pseudo-ternary phase diagrams were prepared to identify the NE region. S_{mix} was evaluated in different weight ratios (1:1, 1:2, 1:3, 2:1, and 3:1). The oil-to- S_{mix} ratios tested included 9:1, 8:2, 7:3, 6:4, 5:5, 4:6, 3:7, 2:8, and 1:9 and were thoroughly mixed prior to titration. At each stage, the mixtures were visually inspected for clarity or turbidity to confirm nanoemulsion formation. Chemix® software was used to construct the pseudo-ternary phase diagrams [12]. The optimized ratios obtained from the diagrams were selected for nanoemulsion formulation.

Preparation of nanoemulsion

The dexlansoprazole nanoemulsion (DEX-NE) containing different Oil/ S_{mix} ratios as mentioned in Table 1, was prepared using a modified method from Sithole et al. (2021) [13]. The drug was first dissolved in the oil phase with continuous vortex mixing at room temperature ($25 \pm 1^\circ\text{C}$). After complete solubilization, Cremophor EL and PEG-300 were added and mixed. Distilled water was then added dropwise to the oil mixture while vortex mixing continued to facilitate emulsification. The resulting coarse emulsion was subjected to magnetic stirring at 1000rpm for 1h, followed by ultrasonication for 10min (three cycles of 2min each) at 80% amplitude. During ultrasonication, the formulation container was immersed in a water or ice bath to dissipate the generated heat and prevent thermal degradation of the formulation.

In vitro characterization

pH measurement

The pH of all formulations was measured using a calibrated pH meter at room temperature.

Phase separation and precipitation assessment

A glass beaker containing 300ml of distilled water and separate beakers containing PBS (pH 6.2 and pH 7.4) were prepared. To each medium, 100 μL of the nanoemulsion formulation was added at room temperature. Samples were observed for phase separation or precipitation at predetermined time intervals (2, 4, 6, 8, 12, and 24h) after storage for 24h [14].

Table 1. Formulation composition of nanoemulsion using different ratios of Oil/ S_{mix} .

Nanoemulsion formulations	Oil/ S_{mix}	S_{mix}	Stirring speed (RPM)
F1	1:9	1:1	1000
F2	1:9	1:2	1000
F3	1:9	1:3	1000
F4	1:9	2:1	1000
F5	1:9	3:1	1000
F6	2:8	1:1	1000
F7	2:8	1:2	1000
F8	2:8	1:3	1000
F9	2:8	2:1	1000
F10	2:8	3:1	1000
F11	3:7	1:1	1000
F12	3:7	1:2	1000
F13	3:7	1:3	1000
F14	3:7	2:1	1000
F15	3:7	3:1	1000
F16	1:9	1:3	800
F17	1:9	1:3	1200
F18	1:9	1:3	1400

Robustness to dilution

The NE was diluted 10-, 50-, 100-, and 200-fold using distilled water, PBS pH 6.2, and PBS pH 7.4 to observe any phase separation or precipitation.

Thermodynamic stability

Formulations were evaluated for physical stability under different stress conditions:

Centrifugation: Samples were centrifuged at 5000rpm for 30minutes.

Heating–Cooling Cycle: Samples were stored for 48h at 40°C , followed by 48h at 4°C .

Freeze–Thaw Cycle: Samples were stored at -20°C and 25°C for 48h at each temperature. One complete cycle constituted one evaluation stage [15,16].

Only physically stable formulations proceeding through all stages without phase separation were selected for further analysis.

The transparency of nanoemulsions was measured using a UV-visible spectrophotometer at 630nm, and results were expressed as % transmittance [17]:

$$\% \text{Transmittance} = T \times 100$$

Where T=Amount of light transmitted (or) Absorbance.

Globule size and zeta potential

The globule size, polydispersity index (PDI), and zeta potential of diluted nanoemulsion (1ml NE diluted 100-fold with distilled water) were measured using a Malvern Zetasizer [18,19].

Drug content

One milliliter of nanoemulsion containing 30mg of DEX was mixed with 5ml of methanol to dissolve the drug. The volume was adjusted to 100ml with PBS (pH 7.4). Following 15min of sonication, the sample was filtered and analyzed using a UV spectrophotometer at 282nm [19].

Entrapment efficiency

Entrapment efficiency (% EE) was determined using the centrifugation method as reported by Rajitha et al. [20]. One milliliter of nanoemulsion was subjected to centrifugation at 15,000rpm for 30min followed by separation of supernatant. The supernatant containing

unentrapped drug was quantified by using UV spectrophotometry at 282nm after proper dilution with methanol. % EE was calculated as:

$$\%EE = \frac{W_T - W_F}{W_T} \times 100$$

W_T : total amount of drug taken

W_F : amount of drug unentrapped

In vitro release studies

In vitro release of DEX from nanoemulsion (DEX-NE) was performed using a dialysis membrane (12–14kDa, HiMedia, India) as reported by Hota et al. [21]. To activate the dialysis membrane, it was first soaked in PBS (pH 6.4) then was filled with 1 ml of DEX-NE (containing 30mg drug) and placed in 50ml of PBS (pH 6.4) maintained at 37°C with continuous stirring at 50rpm. Samples were withdrawn at predetermined intervals (0, 1, 2, 3, 4, 5, and 6h) and replaced with fresh buffer to maintain sink conditions. Absorbance was measured at 282nm using a UV spectrophotometer (Shimadzu UV-2450, Japan).

Ex vivo permeation studies

Ex vivo permeation was performed according to the method reported by Jain et al. [22], with slight modifications, using goat nasal mucosa mounted in a Franz diffusion cell (Zenith, Kolkata, India). Fresh nasal mucosa obtained from a local slaughterhouse was rinsed and stored in PBS (pH 6.4) with aeration until use. The donor compartment contained a formulation equivalent to 30 mg DEX, while the receptor chamber contained 20ml PBS (pH 6.4). The mixture was constantly stirred at 50rpm at 37°C. Samples were withdrawn at 0, 1, 2, 3, and 4 h and replaced with fresh buffer to maintain the sink condition. Absorbance was recorded at 282nm using the UV-spectrophotometer.

Nanoemulsion in situ gel preparation (F3-NEG)

The *in-situ* gel polymer, gellan gum (GG), was gradually dissolved in distilled water heated to $80 \pm 1^\circ\text{C}$ under stirring at 50rpm until complete dissolution of the polymer, followed by cooling to room temperature [23]. The final nanoemulsion (NE), containing 30mg of DEX, was then incorporated into polymeric solutions of varying GG concentrations (0.2–0.5% w/v) and stirred at 50rpm until a homogeneous system was obtained. The prepared nanoemulsion-based *in situ* gel formulations were stored under refrigerated conditions for further evaluation [23].

Evaluation of nanoemulsion in situ gel

1. Gelling capacity

The formulation was mixed with simulated nasal fluid in a 1:1 (v/v) ratio to evaluate its gelation capacity. A magnetic stirrer was used to mix the solution, and the gelation point was identified when gelation caused the magnetic bar to stop moving [24].

2. Clarity Test and pH

Clarity was examined visually against a black-and-white background under adequate lighting conditions [25]. The NEISG was observed for turbidity or the presence of any visible particulate matter. The pH of the DEX nanoemulsion *in situ* gel was measured at room temperature using a calibrated digital pH meter [26].

3. Viscosity studies

Viscosity and rheological properties were measured using a Brookfield digital viscometer (Model RDVT2) [27].

4. Gelation temperature

Gelation temperature was determined by heating 10ml of the formulation placed in a 30ml clear vial initially maintained at a low temperature on the water bath. Then temperature was gradually increased, and the point at which the magnetic bead ceased movement was detected as the gelation temperature. The experiment was performed in triplicate [26].

5. Gelation time

To determine gelation time, 2ml of the *in-situ* gel formulation was transferred to a test tube and incubated in a water bath at $37 \pm 2^\circ\text{C}$. The time at which the formulation no longer flowed upon inversion of the tube was recorded as the gelation time [26].

6. Drug content determination

Precisely, 1 ml of the NE was diluted to 10ml with simulated nasal fluid (SNF). Then, 1 ml of this diluted solution was further diluted to 10ml with SNF for analysis [28].

7. In vitro drug release and ex vivo permeation studies

The same procedures used for the nanoemulsion were applied to estimate the drug release and drug permeability of the *in-situ* gel [12].

8. Drug release kinetics:

The drug release kinetics from the nanoemulsion and nanoemulsion *in-situ* gel were investigated by studying the release data with Zero-order, First-order kinetics, and Higuchi equation. The release mechanism was understood by fitting the data to the Korsmeyer Peppas model.

Stability studies

In any rational design and evaluation of a dosage form for a drug, the stability of the active component must be a major criterion in determining its acceptance or rejection. The purpose of the stability study is to provide evidence on how the quality of a drug substance or drug product varies with time under the influence of environmental factors such as temperature, humidity, and light, enabling recommended storage conditions, retest periods, and shelf-life determination. Stability studies were carried out for the final selected formulation by placing the formulation at ambient temperature ($25 \pm 5^\circ\text{C}$) and refrigerator ($2\text{--}8^\circ\text{C}$) conditions for three months. Nanoemulsions were observed for physical appearance, clarity, pH, and droplet size, 2, 3 months [20,29].

Results

Solubility studies

DEX solubility was evaluated in various oils, surfactants, and co-surfactants. The results revealed that the highest solubility of dexlansoprazole was observed in

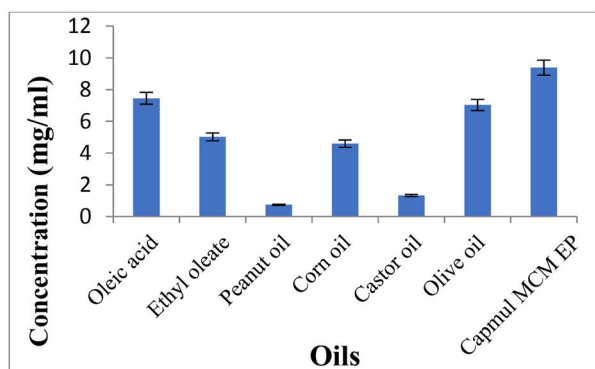
Capmul MCM, Cremophor EL, and PEG 300, as illustrated in Figure 1. Therefore, Capmul MCM, Cremophor EL, and PEG 300 were selected as the oil phase, surfactant, and co-surfactant, respectively, for the nanoemulsion formulation. Similar findings were reported earlier by Kulkarni et al. [30].

Pseudo ternary phase diagrams

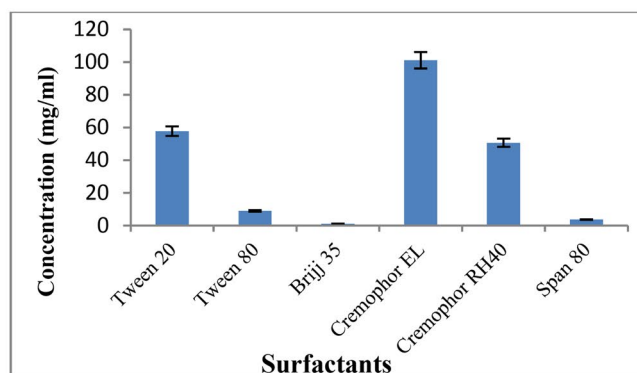
Using Chemix School software, pseudo-ternary phase diagrams were constructed to identify the NE region at varying S_{mix} ratios (3:1, 2:1, 1:1, 1:2, and 1:3) and oil-to- S_{mix} ratios (1:9 to 9:1) (see Supplementary Data). It was observed that increasing the surfactant concentration in the S_{mix} ratios from 1:1 (Figure 2(a)) to 3:1 significantly expanded the NE region, as shown in Figure 2.

The nanoemulsion area expanded with rising amounts of surfactant (Cremophor EL) with relative to the cosurfactants (PEG 300). Blending PEG 300 with Cremophor EL facilitates nanoemulsion dispersion, thereby improving the systemic availability of lipophilic drugs [15].

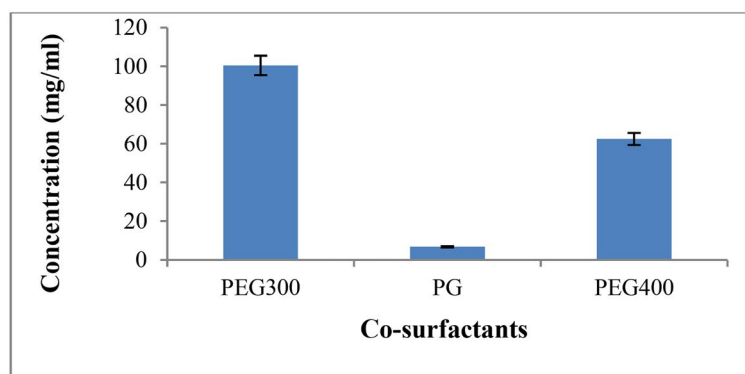
Visual observation of the resulting emulsions during aqueous titration with water was carried out. All the



a)



b)



c)

Figure 1. Solubility of drug in different (a) oils, (b) surfactants, and (c) co-surfactants.

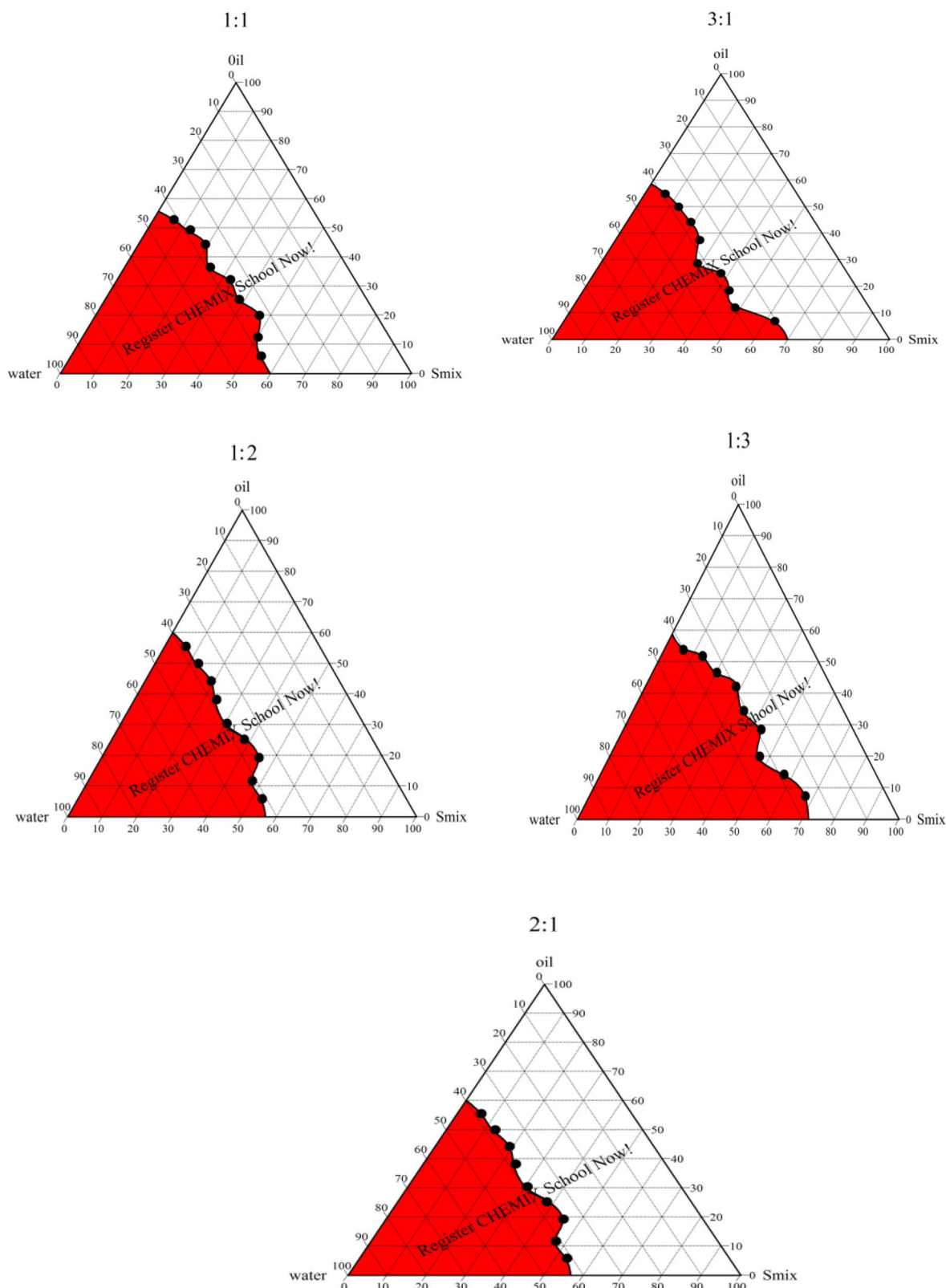


Figure 2. Phase diagram of oil (Capmul MCM EP), S_{mix} (Cremophor EL: PEG 300), and water in the system at various S_{mix} ratios.

formulations dispersed to form either fine or coarse emulsions, which were visually assessed and classified as 'Good' or 'Bad' formulations. Formulations producing transparent, low-viscosity dispersions with good

flowability were considered good nanoemulsions. It was observed that globule size increased with a higher percentage of oil in the S_{mix} , resulting in the formation of hazy dispersions. For oil: S_{mix} ratios of 9:1, 8:2, 7:3,

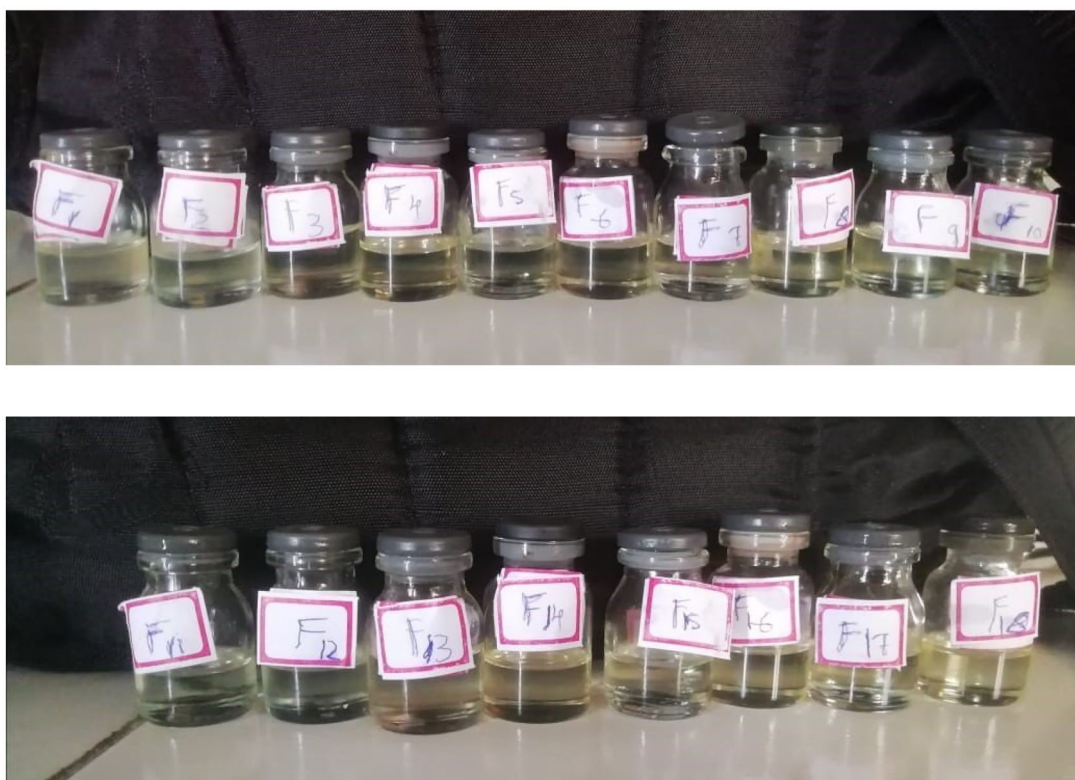


Figure 3. Nanoemulsion formulations.

6:4, 5:5, and 4:6, less clear emulsions were produced and marked as 'B,' whereas clear, transparent emulsions were marked as 'A' (see [Supplementary Data](#)). The S_{mix} ratio of 1:3 exhibited the largest nanoemulsion region, consistent with the findings of Mantena et al. [31]. Based on the pseudo-ternary phase diagrams, the 3:7, 2:8, and 1:9 Oil/ S_{mix} ratios were selected to prepare the various nanoemulsions using 1:1, 1:2, 1:3, 2:1, and 3:1. All the nanoemulsion formulations were transparent, clear, or bluish in appearance, as shown in [Figure 3](#).

In vitro characterization of NE

Phase separation and robustness to dilution. The nanoemulsion was stored for 24h and examined at intervals of 2, 4, 6, 8, 10, and 24h to assess phase separation and drug precipitation. All the NE formulations remained stable and met the evaluation criteria. Additionally, all NEs passed thermodynamic stability studies, and the pH was reported in a range of 5.81 ± 0.01 to 6.52 ± 0.03 which is suitable for the nasal cavity. The % transmittance of DEX-NEs ranged from $89.3 \pm 0.05\%$ to 99.6 ± 0.15 , indicating that the nanoemulsion was transparent and clear, confirming uniform distribution of oil droplets within the continuous phase [32] (Kumar et al. 2018). Viscosity is an important factor affecting the stability and effectiveness of intranasal formulations, as it influences the drug absorption through the nasal mucosa and

residence time in the nasal cavity. The viscosity of NEs ranged from 87.66 ± 0.36 to 111.76 ± 0.15 cps. The drug content of all NEs formulations ranged from $91.06 \pm 0.0019\%$ to $99.06 \pm 0.004\%$.

Globule size, PDI, and zeta potential

Globule size is a key parameter in the characterization of nanoemulsions, as it influences drug release rate, extent of release, and the overall stability of the emulsion. The globule size of the nanoemulsion formulations from F1-F3 and F1, F4, and F5 were found to decrease on increasing the surfactant concentration in S_{mix} ratios for each Oil/ S_{mix} ratio (1:9, 2:8 and 3:7) as shown in [Figure 4](#). Conversely, increasing the percentage of oil in the oil/ S_{mix} ratio (F1, F6, and F11) increased the globule size ([Figure 4](#)). Statistical analysis of the data indicated that the Mean Droplet Size (MDS) is affected by the type and percentage of oil, the surfactant type and the S_{mix} ratio. It was discovered that the MDS increased with increasing the percentage of oil, regardless of the type of oil.

Statistical analysis indicated that the mean droplet size (MDS) is affected by the type and percentage of oil, the surfactant type, and the S_{mix} ratio. It was observed that the MDS increased with increasing oil content, regardless of the oil type.

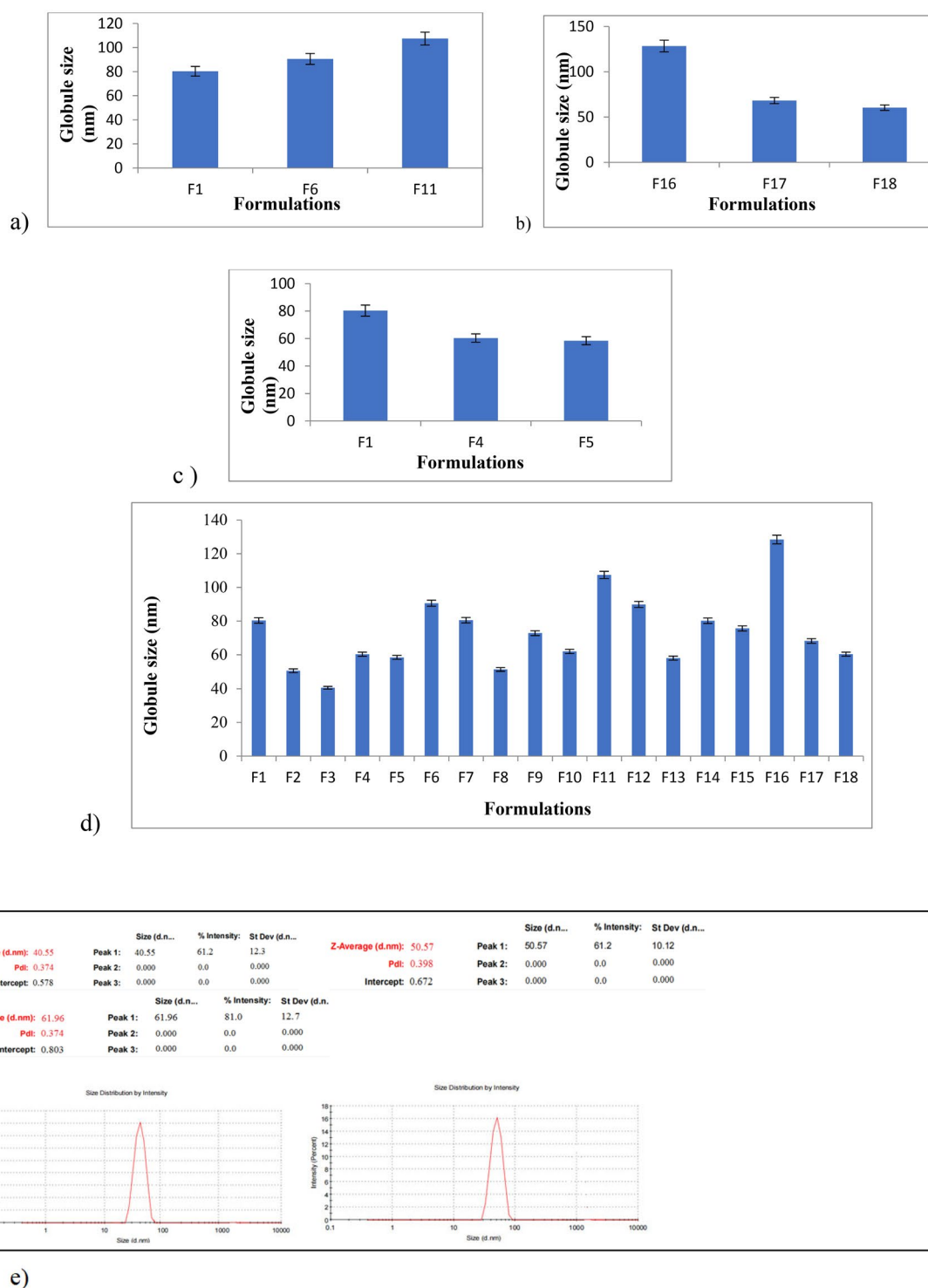


Figure 4. Effect of (a) oil, (b) stirring speed, (c) surfactant on globule size (nm), (d) globule size of nanoemulsion and (e) size distribution of F3, F2, and F10 nanoemulsion.

Nanoemulsions typically have droplet sizes ranging from 20 to 500 nm, as reported by Elsewedy et al. [33], and our formulations (40.55–110.744 nm) fall well within this range. Smaller droplets provide a larger surface area, enhancing their interaction with biological membranes and improving drug absorption. The selected formulation exhibited the smallest droplet

size of 40.55 nm, which is advantageous, as tiny droplets diffuse more rapidly across the mucosa and maintain better physical stability.

The polydispersity index (PDI) indicates the degree of uniformity of droplet size within a nanoemulsion, with values closer to zero representing a narrower and more homogeneous size distribution.

The optimized formulation (F3) exhibited a PDI value of 0.374 (Figure 3). However, it is generally acknowledged that PDI values below 0.3 represent highly monodisperse systems, while PDI values in the range of 0.3–0.5 are widely accepted for nanoemulsions, particularly those prepared using high-energy emulsification methods, where minor droplet size variations are unavoidable without compromising physical stability.

Multiple literature reports clearly establish that nanoemulsions with PDI values well above 0.3 remain homogeneous and stable. Zhang et al. [34] and Penjuri et al. [35] demonstrated that nanoemulsions and self-emulsifying systems with PDI values below 1.0 possess acceptable size distribution and long-term stability. Importantly, Penjuri et al. [35] explicitly reported that formulations with PDI values up to 0.4 showed good homogeneity and no evidence of instability, directly supporting the acceptability of the obtained value.

Further, Nemade et al. [36] and Zanela da Silva Marques et al. [37] reported optimized nanoemulsion systems with PDI values ranging from 0.32 to 0.48, which were considered stable and suitable for drug delivery applications. Similarly, Nair et al. [38] stated that PDI values below 0.5 are indicative of homogeneous nanoemulsion systems with consistently sized droplets. These reports collectively demonstrate that the obtained PDI value of 0.374 lies well within the scientifically accepted range for nanoemulsions.

In the present study, all formulations exhibited PDI values between 0.374 and 0.456. Among these, formulation F3 displayed the lowest PDI, among the tested formulations, indicating the narrowest globule size distribution in a comparative sense. Importantly, this moderate polydispersity was supported by experimental evidence of physical stability, including the absence of phase separation and adequate electrostatic/steric stabilization as reflected by the measured zeta potential. These findings confirm that the observed PDI did not result in globule coalescence or destabilization of the nanoemulsion. Consequently, F3 was rationally selected as the optimized formulation based on comparative evaluation rather than absolute numerical thresholds alone.

Furthermore, as highlighted by Minglani et al. [40], droplet uniformity in nanoemulsions is strongly influenced by the use of appropriate surfactants in combination with high-energy techniques such as probe sonication.

Consistent with this principle, the present formulation employed high-energy sonication along with the nonionic surfactant Cremophor EL and co-surfactant

PEG-300, which together form a robust interfacial film and effectively limit excessive droplet aggregation. This formulation strategy directly supports the observed PDI and confirms that the system possesses acceptable homogeneity and physical stability.

Therefore, the PDI value of 0.374 should be interpreted as indicative of acceptable and literature-supported globule size distribution, rather than poor homogeneity. of acceptable and literature-supported globule size uniformity, and is entirely appropriate for a nanoemulsion developed using high-energy emulsification techniques.

The zeta potential of the optimized formulation (F3) was -17.70 mV. Although higher absolute zeta potential values (± 30 mV) are often associated with electrostatically stabilized colloidal systems, nanoemulsion stability is not governed solely by electrostatic repulsion. In systems stabilized predominantly by nonionic surfactants, steric stabilization plays a dominant role and can effectively prevent droplet coalescence even at relatively low zeta potential values [39–43].

According to Nemade et al. [36], nanoemulsions exhibiting zeta potential values between -15 and $+20$ mV can remain physically stable when appropriate surfactant systems are employed, while Mulam et al. [46] reported that dispersed systems commonly exhibit stability within ± 30 mV depending on the stabilization mechanism involved. In the present study, all DEX nanoemulsion formulations showed zeta potential values between -14.5 and -32.4 mV, and the optimized formulation F3 (-17.70 mV) falls well within the reported stability range.

Importantly, Penjuri et al. [35] reported stable self-emulsifying drug delivery systems with zeta potential values as low as -7.55 to -8.48 mV, demonstrating that high absolute zeta potential values are not mandatory for stability in surfactant-rich systems. Importantly, the negative surface charge of the globules contributes to electrostatic repulsion, which reduces the likelihood of droplet aggregation [47]. However, in the present formulation, stability is predominantly governed by steric hindrance rather than electrostatic repulsion. Nanoemulsions formulated using nonionic surfactants such as Cremophor EL, in combination with co-surfactants like PEG-300, are well documented to remain stable even at low or near-neutral zeta potential values. Such surfactants adsorb at the oil–water interface and form a dense, hydrated, amphiphilic layer that physically prevents close approach of droplets, thereby inhibiting coalescence. Consistent with this mechanism, the obtained zeta potential of -17.70 mV for F3 suggests sufficient interfacial stabilization provided by Cremophor EL and

PEG-300, resulting in effective steric stabilization. Similar observations have been reported for Cremophor-based nanoemulsions exhibiting stable nano-droplet structures despite moderate or low zeta potential values, as also supported by Ahmad et al. [45].

Furthermore, the stability of formulation F3 was experimentally supported by the absence of phase separation, creaming, or droplet growth during the evaluation period, indicating that the combined electrostatic and steric stabilization mechanisms were sufficient to prevent droplet merging and transformation into a coarse emulsion.

Therefore, despite the moderate zeta potential value of -17.70 mV, the optimized formulation is expected to remain physically stable due to the dominant steric stabilization imparted by the nonionic surfactant-co-surfactant system. This confirms that the globules are unlikely to merge and that the nanoemulsion will not undergo destabilization into a coarse emulsion.

For the final selected formulation F3, it was observed that an increase in stirring speed resulted in a decrease

in particle size (F16), whereas a decrease in stirring speed led to an increase in particle size (F17, F18) as shown in Figure 5.

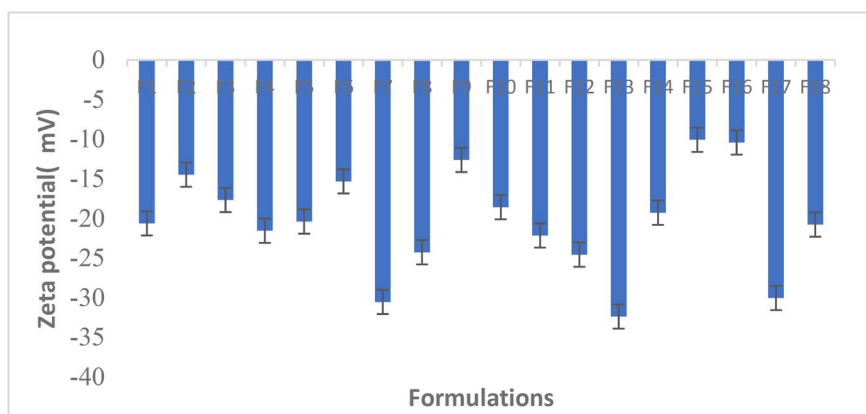
Overall, the combination of small droplet size, low PDI, and appropriate zeta potential contributes to superior nanoemulsion stability, in alignment with previous studies.

Entrapment efficacy

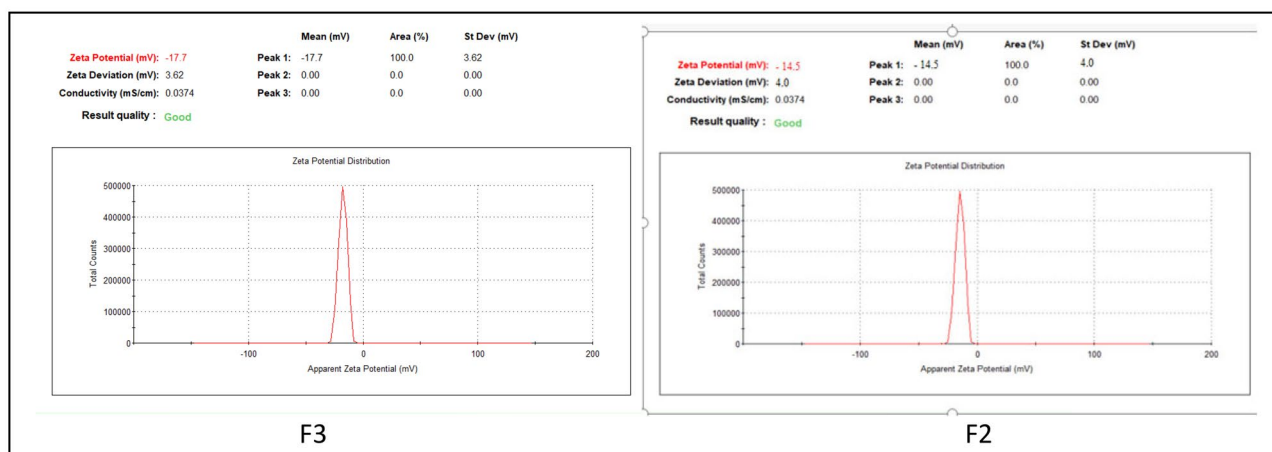
All the NE formulations exhibited high entrapment efficiency, ranging from 82–95%. The high %EE indicates effective solubilization of dexlansoprazole within the oil phase and stabilization by the surfactant-co-surfactant system. Our results are in agreement with the findings of Jadhav et al. [48].

In vitro drug release

The dialysis bag method was used to perform the *in vitro* drug release profile, and the formulations were compared with the API suspension. As shown in Figure 6, the



a)



b)

Figure 5. (a) Zeta potential of Nanoemulsion formulations (b) Zeta potential curve of F3 and F2 formulations.

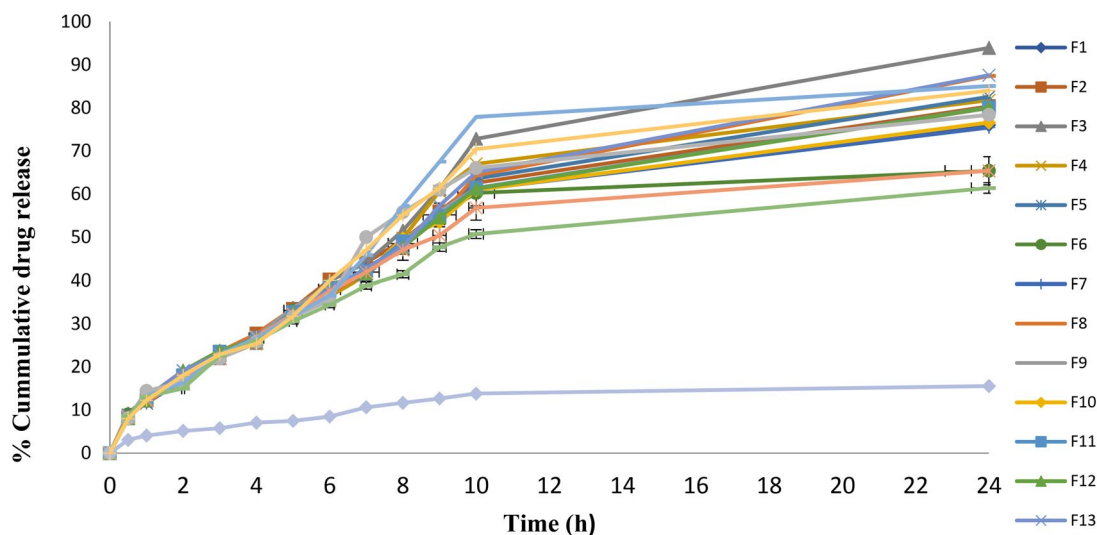


Figure 6. *In vitro* drug release of all nanoemulsion formulations up to 24 h.

cumulative release of DEX from NEs was significantly higher compared to API suspension in PBS pH 6.4, likely due to the higher solubility of the drug [18]. The pure drug suspension demonstrated only $10.32 \pm 0.56\%$ drug release in PBS pH (6.4) after 8 h whereas all NEs exhibited significantly higher drug release, ranging from 47.04 ± 0.075 to $55.82 \pm 0.059\%$. Among all formulations, F3 shows the highest drug release i.e. $55.82 \pm 0.059\%$ at 8 h.

At the end of 24 h, the pure drug suspension achieved only $22.59 \pm 0.49\%$ drug release, while F3 depicted $93.93 \pm 0.031\%$ release. This enhanced release profile is consistent with previous studies, as reported by Jaber and Rajab [49], who observed faster dissolution and higher release rates from nanoemulsions compared with pure drug suspensions. The nano-sized droplets provide a larger interfacial area and improved wettability, which together accelerate diffusion into the release medium.

This higher oil percentage may contribute to a retardation in the release from the oil droplet into the surrounding aqueous phase since the presence of higher oil (at constant surfactant concentration) will result in the formation of larger droplets, leading to a reduction in the surface area with a reduced rate of diffusion.

In brief, nanoemulsion formulations were developed by systematically varying the oil concentration and S_{mix} ratio. All batches were initially screened for visual appearance and physical stability, followed by detailed evaluation based on critical quality attributes, including mean droplet size, polydispersity index (PDI), zeta potential, drug content, and *in vitro* drug release. Formulation F3 was selected as the optimized nanoemulsion due to its superior overall performance, characterized by nanoscale droplet size (40.55 nm),

moderately narrow and acceptable size distribution (PDI 0.374), adequate electrokinetic stability (-17.70 mV), and the highest *in vitro* drug release ($93.93 \pm 0.031\%$ at 24 h).

The optimized formulation was selected based on achieving an optimal balance among these parameters. Specifically, preference was given to a formulation exhibiting a nanoscale droplet size with a narrow size distribution, sufficient electrokinetic stability, and higher *in vitro* drug release.

Subsequently, F3 was incorporated into gellan gum-based *in situ* gels and further optimized by varying the polymer concentration.

TEM

TEM analysis was performed for the final F3 formulation to evaluate its morphological characteristics, and the results are displayed in Figure 7. The TEM images revealed that the nanoemulsions are spherical in shape and have a nanometric range. Similarly, spherical morphology has been consistently reported in nanoemulsions studies by Lin et al. [50], confirming the integrity and uniformity of the droplets. The observed spherical morphology further supports stability by reducing aggregation and ensuring consistent drug release.

Formulation and evaluation of DEX nanoemulsion in situ gel

Different *in situ gel* formulations containing F3 nanoemulsion were prepared using varying concentrations of gellan gum (0.2–0.6%) as an ion-triggered gelling agent. These formulations were evaluated for

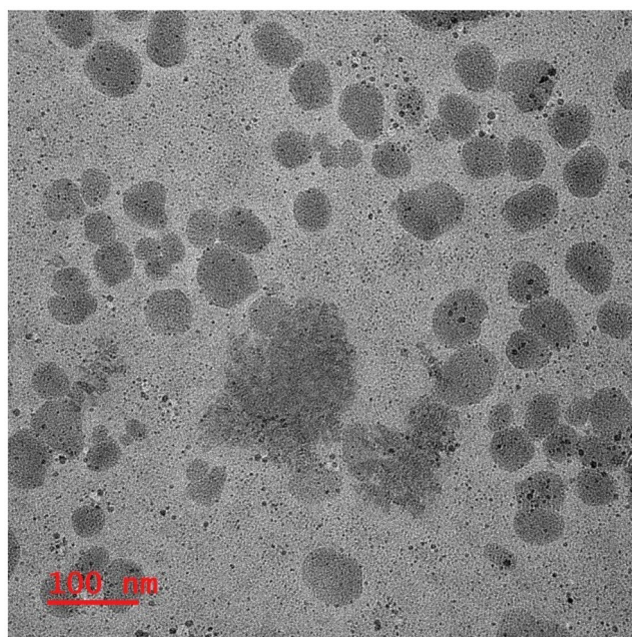


Figure 7. Transmission electron microscopic image of F3 formulation.

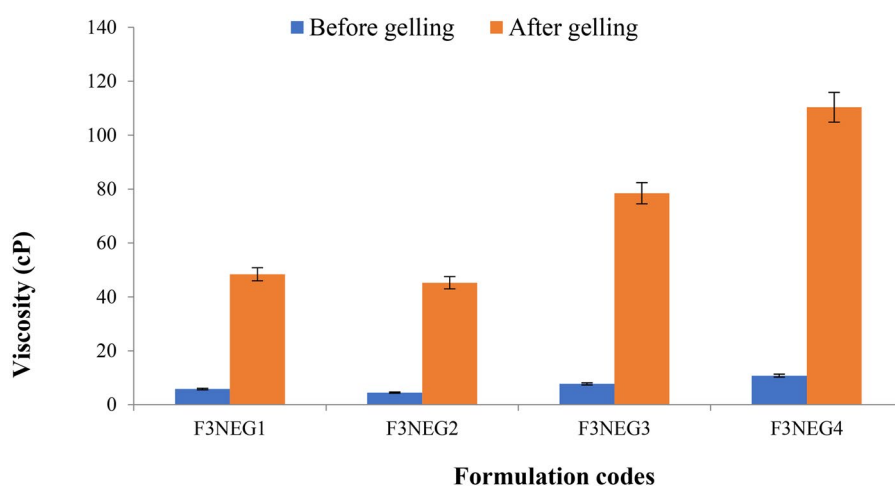


Figure 8. Viscosity of nanoemulsion *in situ* gel (NEGs) before and after gelling.

gelling capacity, clarity, pH, viscosity, gelation temperature, gelling time, syringeability, spreadability, drug content, and *in vitro* drug release.

Gelling capacity, clarity, pH and viscosity of *in situ* gel formulations

It was observed that gel intensity increased with higher polymer concentrations. The results show that all formulations could induce gelation, as reported by Ji et al. [51]. At lower polymer concentrations, gelling took a few minutes and gels could be redispersed, whereas at higher concentrations (0.6% gellan gum), gelling occurred immediately and remained stable for several hours.

All formulations were transparent, with no particulate matter observed, and successfully passed the clarity test. The pH of all F3NEG formulations ranged from 5.63 ± 0.0131 to 6.01 ± 0.0147 at 25°C [52]. Viscosities of the *in-situ* gel solutions, both before and after gelation, were determined using Brookfield viscometer results presented in Figure 8. An increase in polymer concentration led to a corresponding increase in viscosity. The highest viscosity (110.346 ± 0.181 cps) was observed with F3NEG4 at a polymer concentration of 0.6% w/v, while the lowest viscosity (48.354 ± 0.145 cps) was recorded for F3NEG1 at 0.2% w/v. These findings are consistent with previous intranasal *in-situ* gel studies, where increasing gellan gum concentrations enhanced gel strength, mucoadhesion, and residence time as suggested by Salunke and Patil [53].

Gelation temperature and time

The gelation temperature and gelling time of all F3NEG formulations were in the ranges of $34.1 \pm 0.175^\circ\text{C}$ to $40.3 \pm 0.125^\circ\text{C}$ and 62 to 120 s, respectively, as displayed in Table 2 and Figure 9. The shortest gelling time was recorded with the F3NEG4 formulation, which contained 0.5% polymer.

The gellan sol-gel transition is an exothermic process that can occur in the range of temperatures between 30° and 50°C , depending on the specific nature of the cation, polymer concentration, and presence of co-solutes [54]. These results are consistent with previous studies on gellan gum-based intranasal gels, which reported gelation temperatures between 30° and 40°C and rapid gelation (60–180 s) upon exposure to simulated nasal ion content [55]. The temperature-triggered sol-gel transition is advantageous for nasal delivery, ensuring immediate gelation once the formulation contacts nasal fluids, thus enhancing residence time and drug absorption.

Percentage drug content of F3NEG

All NE *in situ gel* formulations exhibited drug contents ranging from $98.54 \pm 0.062\%$ to $99.65 \pm 0.062\%$, as

Table 2. Gelling temperature, gelling time and drug content of different NE *in situ gel* formulations ($n=3$).

Formulations Codes	Gelling temperature	Gelling time (S)	% Drug content (AM $\pm$ SD)
F3NEG 1	40.3 ± 0.125	120 ± 0.114	98.54 ± 0.072
F3NEG 2	37.15 ± 0.148	103 ± 0.075	99.06 ± 0.051
F3NEG3	35.2 ± 0.117	87 ± 0.064	99.68 ± 0.145
F3NEG4	34.1 ± 0.175	62 ± 0.123	99.65 ± 0.062

shown in Table 2. High drug content ($>98\%$) indicates uniform drug distribution throughout the gel matrix and efficient incorporation of the nanoemulsion droplets. These findings are consistent with previous reports on gellan gum-based *in situ* gels, which achieved drug contents greater than 95% for mucosal delivery systems [53].

In vitro release studies

The *in vitro* drug release profile for all F3NEG formulations was evaluated using the dialysis bag method and the results are presented in Figure 10. The results indicated that the NE *in situ* gels exhibited higher drug release compared to the drug suspension. Among all formulations, F3NEG4 exhibited the highest drug release, reaching 86.49% after 24 h. This enhanced release is attributed to the improved solubility and dissolution rate of DEX.

Similar improvements in drug release have been reported for nanoemulsion-loaded nasal gels, where the small droplet size and enhanced wettability significantly increase dissolution and diffusion rates [56]. Nanoemulsion-based delivery systems further improve drug dissolution by reducing interfacial tension and providing a large surface area for diffusion.

The ability of nanoemulsion *in situ gel* formulations to enhance bioavailability has been well documented. Earlier studies have shown that nanoemulsion gels significantly improve mucosal absorption and systemic availability of poorly soluble drugs delivered via the nasal route [57].

Among the tested F3NEG formulations, F3NEG4 was selected as the optimized batch based on its overall

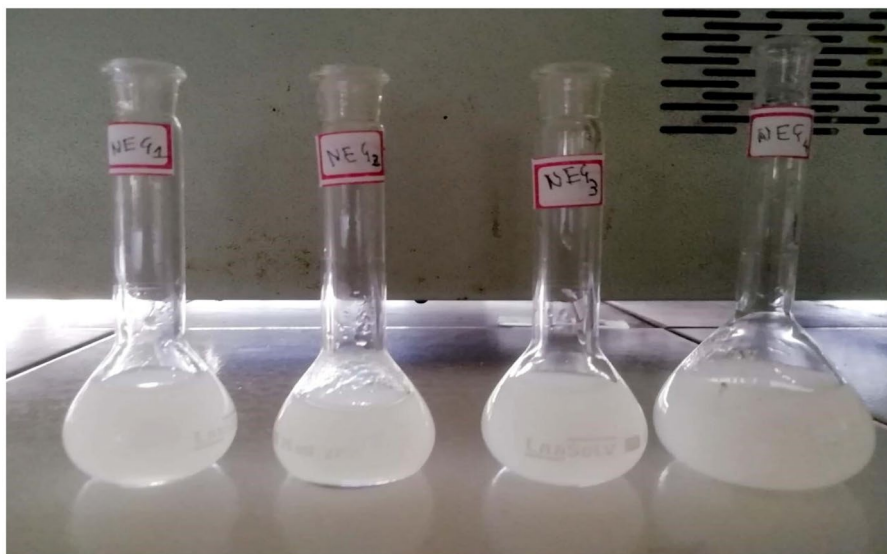


Figure 9. Nanoemulsion *in situ gel* formulation after gelling.

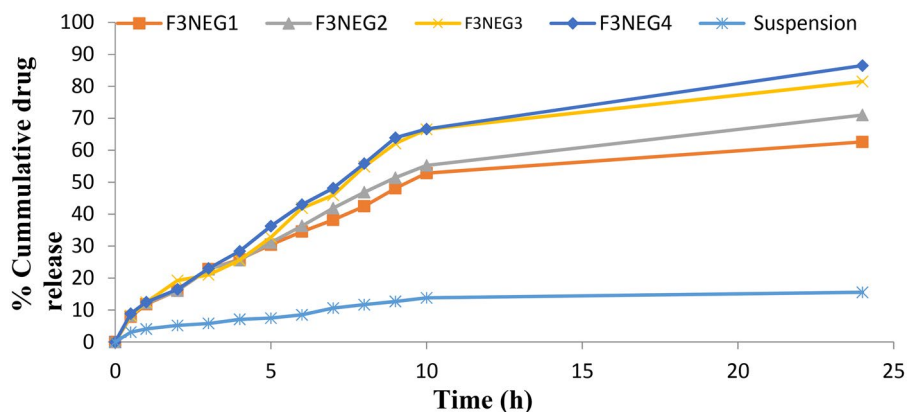


Figure 10. *In vitro* drug release from nanoemulsion *in situ* gel formulations.

superior performance. It exhibited the shortest gelation time (62 s), appropriate gelling strength and capacity, acceptable drug content, and significantly higher *in vitro* drug release (86.49% at 24 h). The selection was based on a comprehensive, comparative evaluation of critical parameters, including gelling time, gelling strength, gelling capacity, drug content, and release behavior. The formulation that best satisfied these collective criteria was designated as the optimized formulation.

Overall, the optimized formulation was selected based on a comparative, multi-parameter evaluation, rather than a single attribute, ensuring a balanced performance in terms of stability, release behavior, and suitability for intranasal administration.

Ex vivo permeation studies

The F3NE and F3NEG4 formulations exhibited drug permeation of $90.02 \pm 0.565\%$ and $80.688 \pm 0.057\%$, respectively, after 24 h. DEX-NE showed enhanced permeability compared to the DEX suspension, as illustrated in Figure 11. Drug permeation across goat nasal mucosa was significantly higher for the NE and NE *in situ* gel formulations than for the drug suspension. Among the *in-situ* gel formulations, F3NEG4 demonstrated the highest drug permeation. Using nanoemulsion systems also offers the advantage of increased drug absorption via the nasal route by providing a larger surface area for contact with epithelial cells.

This enhanced permeation can be attributed to the ultra-small droplet size of nanoemulsion, which increases surface area, improves drug solubilization, and facilitates tight epithelial junction [56,58]. Additionally, the *in-situ* gel matrix prolongs mucosal residence time, allowing sustained contact between drug-loaded droplets and the epithelium, thereby increasing drug flux in nasal delivery systems [57].

Drug release kinetics

Drug release data of the selected formulations F3 and F3NEG4 were analyzed using various kinetic models, including Zero-order, First-order, Higuchi, Hixson–Crowell, and Korsmeyer–Peppas models, with the best fit determined based on the highest coefficient of determination (R^2) as mentioned in Table 3. Among these, the Higuchi model exhibited the highest R^2 values for F3 (0.9892) and F3NEG4 (0.9598), indicating that drug release from both formulations is predominantly diffusion-controlled.

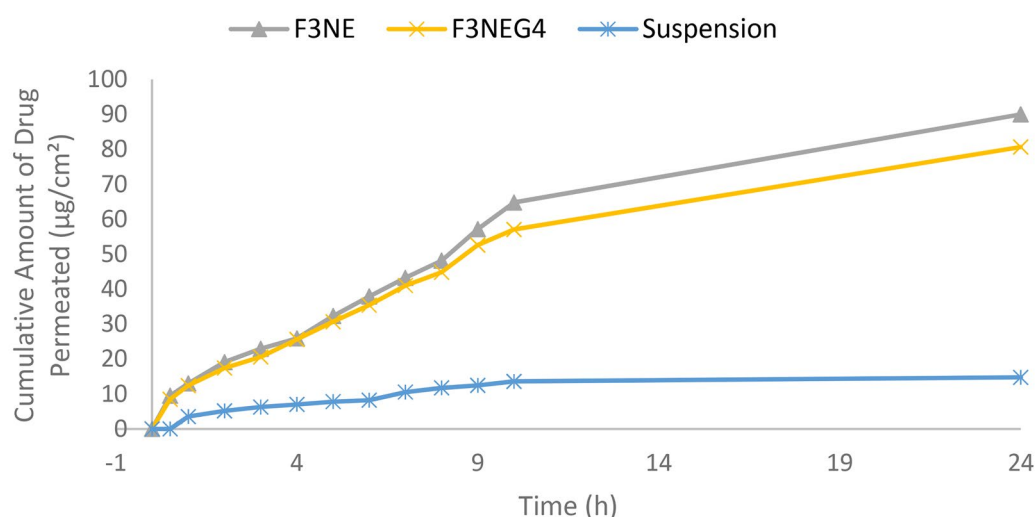
Compliance with the Higuchi model is particularly beneficial in the context of the present research objective, which aims to achieve prolonged and controlled drug delivery via the intranasal route. Diffusion-controlled release from a matrix system ensures gradual drug transport driven by a concentration gradient, thereby minimizing initial burst release and maintaining sustained drug availability at the nasal mucosa. Such a release profile is desirable for intranasal delivery, as it supports prolonged local residence time and consistent absorption while reducing the need for frequent dosing.

Further insight into the release mechanism was obtained from the Korsmeyer–Peppas model, which yielded release exponent (n) values of 0.581 and 0.574 for F3 and F3NEG4, respectively, indicating a non-Fickian (anomalous) release mechanism. This behavior reflects the combined contribution of drug diffusion and polymer relaxation or swelling, confirming the functional role of the gellan gum-based *in situ* gel system. Upon exposure to nasal ionic conditions, the formulation undergoes sol–gel transition, forming a viscoelastic gel matrix that modulates drug diffusion while resisting rapid clearance from the nasal cavity. Similar findings were reported by Chime et al. [59].

The coexistence of diffusion-controlled (Higuchi) and polymer-relaxation-controlled (non-Fickian) release



a)



b)

Figure 11. (a) Goat nasal mucosa and complete set of Franz diffusion cell with goat nasal mucosa, (b) *Ex vivo* permeation studies using nasal goat mucosa in Franz diffusion cell up to 24 h.

Table 3. Release kinetics of F3 and F3NEG4.

Formulation	Zero-order R^2	First-order R^2	Higuchi R^2	Hixson-Crowell R^2	Korsmeyer-Peppas R^2	Higuchi n
F3	0.9293	0.9797	0.9892	0.5466	0.9490	0.581
F3NEG4	0.8434	0.9401	0.9598	0.5083	0.9084	0.574

is advantageous for nanoemulsion-based *in situ* gels, as drug release is regulated both by diffusion from nanoemulsion droplets and by the dynamic gel network formed *in situ*. Similar release behavior has been reported for nanoemulsion-based *in situ* gel systems by Vijaya Rani et al. [25]. Thereby, highlighting their suitability for sustained intranasal drug delivery.

Overall, the predominance of Higuchi kinetics in combination with a non-Fickian release mechanism demonstrates that the developed nanoemulsion-based *in situ* gel effectively controls drug diffusion following gelation in the nasal environment. This controlled and

sustained release behavior directly aligns with the study objective of enhancing nasal residence time and achieving prolonged drug delivery via the intranasal route.

Stability studies

Stability studies were conducted at room temperature and refrigerated temperature for three months and were evaluated for physical appearance, clarity test, precipitation and phase separation, and globule size. The nanoemulsion showed no signs of phase

separation, precipitation, or loss of clarity throughout the study. The pH and mean globule size were nearly unchanged after three months. Overall, the formulation maintained its physicochemical stability throughout the study. Stability studies were performed on the final nanoemulsion, as it serves as the primary carrier system.

Discussion

The development of a nanoemulsion-based *in situ* gel for the nasal delivery of Dexlansoprazole aims to address the major challenges associated with its conventional oral administration, namely poor bioavailability primarily due to its poor aqueous solubility belonging to BCS class II, extensive first-pass metabolism and its acid-labile nature in the gastric environment. These limitations significantly reduce the therapeutic effectiveness of Dexlansoprazole when administered orally.

To overcome the aforementioned limitations, intranasal delivery using a nanoemulsion-based *in situ* gel offers several advantages, such as avoiding hepatic first-pass metabolism, enhancing solubility, prolonging drug release, providing a rapid onset of action, and enhancing patient compliance through its noninvasive route and improving bioavailability [60].

Nanoemulsions, being thermodynamically stable dispersions with droplet sizes typically in the nanometer range, provide several benefits such as enhanced solubility of lipophilic drugs, stable drug dispersion, improved absorption, and faster onset of action [61]. Their small droplet size increases the surface area for drug absorption across the nasal mucosa. Additionally, the intranasal route bypasses the GIT and liver metabolism, offering a direct path to the systemic circulation [62].

Incorporating the nanoemulsion into an *in-situ* gel enhances the formulation's nasal residence time. Polymers such as Poloxamer 407 (temperature-sensitive), Carbopol 934P (pH-sensitive), and gellan gum are used to achieve sol-to-gel transition at physiological nasal conditions. This gelation mechanism ensures that the formulation remains in place post-administration, minimizing mucociliary clearance, thereby improving residence time, mucosal adhesion, and enabling sustained drug release [63].

Preformulation studies were carried out to support the development of the nanoemulsion-based *in situ* gel formulation. Solubility studies were performed to identify suitable oils, surfactants, and co-surfactants capable of effectively dissolving Dexlansoprazole. The screening results revealed that Dexlansoprazole

exhibited the highest solubility in Capmul MCM as an oil, Cremophor EL as a surfactant, and PEG 300 as a co-surfactant. These components were therefore selected as an oil phase, surfactant, and co-surfactant for the nanoemulsion formulation. The addition of appropriate co-surfactants reduces the interfacial tension between the two phases, facilitating the formation of a stable nanoemulsion (Kumar et al. 2018) [32].

The nanoemulsion was prepared using high-speed homogenization followed by ultrasonication to achieve a uniform and stable formulation. To identify the suitable nanoemulsion region, pseudo-ternary phase diagrams were constructed using various ratios of S_{mix} (a combination of surfactant and co-surfactant). The nanoemulsion area expanded with increasing amounts of surfactant (Cremophor EL) relative to the cosurfactants (PEG 300). The blending of PEG 300 with Cremophor EL facilitates the dispersion of nanoemulsion [64]. The 1:3 ratio displayed the highest nanoemulsion region.

All NE preparations passed the thermodynamic stability studies, and the pH was within the nasal range. Percent transmittance results of DEX demonstrated that the formulations were transparent and clear, confirming the uniform distribution of oil droplets within the continuous phase [32] (Kumar et al. 2018).

Globule size is a key parameter in the characterization of nanoemulsions, as it influences the drug release rate, extent of release, and the overall stability of the emulsion [65]. Statistical analysis indicated that Mean Droplet Size (MDS) is affected by the type and percentage of oil, the surfactant type and the S_{mix} ratio. It was observed that the MDS increased with increasing the oil percentage, regardless of the oil type. The smallest globule size was observed in the F3 formulation. The nanoemulsion exhibited a droplet size below 129 nm, promoting better nasal absorption.

Additionally, the slight increase in droplet size at higher oil concentrations led to a reduced drug release rate, highlighting that smaller globules do not always correlate with faster release, as suggested by Sandri et al. [66]. This finding emphasizes the importance of formulation-specific optimization and reinforces the role of nanoemulsions in achieving controlled drug release. The optimized formulation (F3) exhibited a PDI value of 0.374 (Figure 3). However, it is generally acknowledged that PDI values below 0.3 represent highly monodisperse systems, while PDI values in the range of 0.3–0.5 are widely accepted for nanoemulsions, particularly those prepared using high-energy emulsification methods, where minor droplet size variations are unavoidable without compromising physical stability.

Multiple literature reports clearly establish that nanoemulsions with PDI values well above 0.3 remain homogeneous and stable. Literature reports confirm that nanoemulsions with PDI values below 1.0 possess adequate size uniformity and physical stability [34,35]. Notably, Penjuri et al. reported nanoemulsions with PDI values up to 0.4 showed good homogeneity and no evidence of instability, directly supporting the acceptability of the obtained value. Similarly, Nemade et al. [36] and Zanela da Silva Marques et al. [37] reported optimized nanoemulsions with PDI values ranging from 0.32 to 0.48, while Nair et al. [38] stated that PDI values below 0.5 indicate homogeneous nanoemulsion systems. These reports collectively demonstrate that the obtained PDI value of 0.374 lies well within the scientifically accepted range for nanoemulsions.

All formulations showed PDI values between 0.374 and 0.456, with formulation F3 exhibiting the lowest PDI, among the tested formulations, indicating the narrowest globule size distribution in a comparative sense. The acceptable PDI was supported by the observed physical stability, evidenced by the absence of phase separation and adequate electrostatic/steric stabilization as reflected by the zeta potential, confirming that moderate polydispersity did not adversely affect stability. Consequently, F3 was rationally selected as the optimized formulation based on comparative evaluation rather than absolute numerical thresholds alone. The use of high-energy sonication in combination with Cremophor EL and PEG-300 further contributed to effective interfacial stabilization and droplet uniformity, justifying the selection of F3 as the optimized formulation [40].

Consistent with this principle, the present formulation employed high-energy sonication along with the nonionic surfactant Cremophor EL and co-surfactant PEG-300, which together form a robust interfacial film and effectively limit excessive droplet aggregation. This formulation strategy directly supports the observed PDI and confirms that the system possesses acceptable homogeneity and physical stability.

The zeta potential ensured the electrostatic stability of the formulation [67]. It was reported that the zeta potential value of nanoemulsion lies in the range of ± 40 mV, showing higher stability (Fanyu et al. 2021) [46]. The optimized formulation (F3) exhibited a zeta potential of -17.70 mV. Although higher absolute zeta potential values (± 30 mV) are often associated with electrostatically stabilized colloidal systems, nanoemulsion stability is not governed solely by electrostatic repulsion. In systems stabilized predominantly by nonionic surfactants, steric stabilization plays a dominant role and can effectively prevent droplet coalescence even at relatively low zeta potential values [39–43]

Literature reports indicate that nanoemulsions with zeta potential values between -15 and $+20$ mV can remain physically stable when appropriate surfactant systems are employed [36], and stable surfactant-rich systems have been reported even at lower values [35,46]. In the present study, all formulations showed zeta potential values between -14.5 and -32.4 mV, with F3 falling well within the reported stability range. In the present study, all DEX nanoemulsion formulations showed zeta potential values between -14.5 and -32.4 mV, and the optimized formulation F3 (-17.70 mV) falls well within the reported stability range.

The negative surface charge contributes to electrostatic repulsion; however, stability in F3 (-17.70 mV) is mainly attributed to steric hindrance provided by Cremophor EL and PEG-300. These nonionic surfactants adsorb at the oil–water interface and form a dense, hydrated amphiphilic interfacial layer around the droplets, effectively preventing close approach and coalescence even at moderate zeta potential values. Similar observations have been reported for Cremophor-based nanoemulsions exhibiting stable nano-droplet structures despite moderate or low zeta potential values, as also supported by Ahmad et al. [45].

Furthermore, the physical stability of formulation F3 was confirmed by the absence of phase separation, creaming, or droplet size increase throughout the study period. These observations indicate that the combined electrostatic and steric stabilization effectively inhibited droplet coalescence. Although the zeta potential was moderate (-17.70 mV), the strong steric barrier provided by the nonionic surfactant-co-surfactant system is expected to ensure sustained stability, preventing droplet aggregation and conversion into a coarse emulsion.

Transmission Electron Microscopy (TEM) confirmed the spherical shape, without aggregation, and the results are in line with earlier studies [50].

In vitro drug release was conducted using a dialysis membrane in phosphate buffer saline (PBS) at pH 6.4. The results revealed that the cumulative release of DEX from nanoemulsion (NEs) was significantly higher compared to the drug suspension. Among all NEs, F3 shows the highest drug release, i.e. $55.82 \pm 0.059\%$ at 8 h and significantly higher than that of the pure drug suspension ($10.32 \pm 0.56\%$). F3 exhibited $93.93 \pm 0.031\%$ drug release at the end of 24 h. A sustained release of the drug was observed over 24 h, which was attributed to the gel matrix entrapment and controlled diffusion of the drug.

Formulation F3 was selected as the optimized nanoemulsion due to its superior overall performance, characterized by nanoscale droplet size (40.55 nm), moderately narrow and acceptable size distribution (PDI

0.374), adequate electrokinetic stability (-17.70 mV), and the highest *in vitro* drug release ($93.93 \pm 0.031\%$ at 24 h).

The optimized formulation was selected based on achieving an optimal balance among these parameters. Specifically, preference was given to a formulation exhibiting a nanoscale droplet size with a narrow size distribution, sufficient electrokinetic stability, and higher *in vitro* drug release.

The ability of nanoemulsion *in situ* gel formulations to enhance bioavailability has been well documented. Earlier studies showed that nanoemulsion gels significantly improve mucosal absorption and systemic availability of poorly soluble drugs delivered *via* the nasal route [57,68]. A higher oil percentage may contribute to retardation in the release from the oil droplet into the surrounding aqueous phase, as the presence of more oil (at constant surfactant concentration) leads to the formation of larger droplets, reducing the surface area and decreasing the rate of diffusion [69].

The selected nanoemulsion exhibited a markedly reduced droplet size and favorable interfacial properties, which correlated with enhanced solubility, *in vitro* release, and *ex vivo* permeation of DEX. These findings suggest the formulation has the potential to strengthen bioavailability upon nasal administration. Future *in vivo* studies are required to confirm these outcomes. Thus, the potential improvement in bioavailability remains indicative rather than conclusive.

Subsequently, formulation F3 was incorporated into gellan gum-based *in situ* gelling systems and optimized by adjusting the polymer concentration. Gellan gum was added at concentrations ranging from 0.2 to 0.6% to enable sol-gel transition upon contact with nasal fluid. The final formulations were thoroughly evaluated for their gelling capacity, viscosity, and gelation temperature to ensure they met the required criteria for effective nasal delivery [70].

The *in-situ* gel exhibited a pH compatible with the nasal mucosa, ranging from 5.5 to 6.5. It demonstrated a rapid sol-to-gel transition within seconds upon contact with nasal fluid. The gel strength was sufficient to prevent drainage, while the viscosity was low during administration but increased post-gelation to ensure better retention. Additionally, the mucoadhesive strength of the gel prolonged the residence time in the nasal cavity. The shortest gelling time was recorded for F3 NEG4 formulation with a polymer concentration of 0.5%.

Results of *in vitro* drug release indicated that NE *in situ* gel formulations (F3NEG) achieved higher drug release compared to the drug suspension. Among these, F3NEG4 exhibited the highest drug release of 86.49% after 24 h. This enhanced release is attributed to the improved solubility and dissolution rate of DEX.

The ability of nanoemulsion-based *in situ* gel to enhance bioavailability has been previously reported, primarily through improved mucosal absorption, prolonged residence time, and enhanced permeability [38,39].

Among the tested models, the Higuchi model exhibited the highest R^2 values for F3 (0.9892) and F3NEG4 (0.9598), indicating that drug release from both formulations is predominantly diffusion-controlled. This release behavior is advantageous for the present intranasal delivery objective, as diffusion-controlled release from a matrix system minimizes burst release and ensures sustained drug availability at the nasal mucosa, thereby supporting prolonged residence time and consistent absorption while reducing the need for frequent dosing.

Further elucidation of the release mechanism using the Korsmeyer-Peppas model revealed n values of 0.581 for F3 and 0.574 for F3NEG4, indicative of anomalous (non-Fickian) transport involving a combination of drug diffusion and polymer relaxation/swelling. This release behavior highlights the functional contribution of the gellan gum-based *in situ* gel, which undergoes ion-triggered sol-gel transition in the nasal environment to form a viscoelastic network that regulates drug diffusion and enhances resistance to mucociliary clearance [54,55]. The coexistence of diffusion-controlled (Higuchi) and polymer-relaxation-controlled (non-Fickian) release is particularly advantageous for nanoemulsion-based *in situ* gels, as both nanoemulsion droplets and the dynamic gel network regulate drug release. Similar release behavior has been reported for nanoemulsion-based *in situ* gel systems by Vijay Rani et al. [55]. Thereby, highlighting their suitability for sustained intranasal drug delivery. Overall, these release characteristics directly aligns with the study objective of enhancing nasal residence time and achieving prolonged drug delivery via the intranasal route [71].

Ex vivo studies were conducted using freshly excised goat nasal mucosa mounted on Franz diffusion cells. The cumulative drug permeation was significantly higher for the nanoemulsion *in situ gel* compared to the control drug suspension. DEX-NE showed enhanced permeability compared to DEX suspension, demonstrating the potential of the formulation to improve nasal drug absorption.

Among the NEG formulations, F3NEG4 displayed the highest drug permeation, with more than 80% drug permeation recorded, whereas only approximately 20% was observed with the drug suspension. This enhancement may be attributed to the nanoemulsion's ultra-small droplet size, which increases drug solubilization, prolongs residence time, and improves permeability through the

nasal mucosa. Utilizing nanoemulsion systems offers the additional benefit of improving the surface area for drug-epithelial cell contact, thereby facilitating enhanced absorption via the nasal route.

The final formulation, F3NEG4, exhibited superior drug release (86.49% after 24 h), excellent permeability in *ex vivo* studies (80.68%), and stability under storage conditions. The findings confirm that the nanoemulsion approach significantly improves drug dissolution compared to conventional drug suspension.

The results align with previous studies on nanoemulsion-based drug delivery systems, particularly in enhancing drug solubility and permeability [6,32]. The use of gellan gum as an *in-situ* gelling polymer further supports prior research indicating its efficacy in prolonging drug residence time in nasal applications [7]. Moreover, the study contributes to the growing body of work advocating nasal drug delivery as an effective alternative to bypass first-pass metabolism and improve systemic availability [4].

An unexpected finding in this study was the significantly higher drug permeation observed in F3NEG4 compared to the drug suspension and other *nanoemulsion in situ* gel formulations. This improvement may be attributed to the optimal balance of surfactant and co-surfactant ratios, which facilitated superior drug solubilization and membrane permeability.

The nanoemulsion *in situ* gel of Dexlansoprazole for intranasal delivery demonstrates enhanced drug solubility and stability, effective nasal retention, and improved mucosal permeability. It also provides sustained release [72]. These findings support the potential of intranasal nanoemulsion-based *in situ* gel systems as a promising strategy for the systemic delivery of acid-labile drugs like Dexlansoprazole.

This study lays the groundwork for further innovations in noninvasive drug delivery systems. The findings hold implications for expanding the application of nanoemulsion *in situ* gels for other hydrophobic drugs, potentially revolutionizing drug administration for various therapeutic areas.

Conclusion

In conclusion, the developed nanoemulsion (NE) and gellan gum-based nanoemulsion *in situ* gelling systems of dexlansoprazole demonstrated prolonged nasal residence time and sustained drug release *in vitro*, along with favorable *ex vivo* permeation characteristics. These findings indicate the potential of the intranasal delivery system to improve drug absorption and support enhanced systemic availability. Among the investigated nanoemulsion *in situ* gel formulations,

F3NEG4 emerged as the optimized formulation based on its optimal gelation behavior, drug release profile, and superior *ex vivo* permeation performance. Therefore, the nanoemulsion-based *in situ* gel may have the potential to enhance bioavailability based on the improved solubility, dissolution and *in vitro* characteristics observed; however, this requires confirmation through *in vivo* studies. Accordingly, the suggested improvement in bioavailability remains indicative rather than conclusive.

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Author contributions

CRedit: **Gyati Shilakari**: Conceptualization, Formal analysis, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – review & editing; **Sai Kumar Uppala**: Writing and editing; **Meghamala G.**: Writing – review & editing; **Srilatha Bhukya**: editing, correction and formatting; **Abhay Asthana**: Resources, Validation.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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